# BIAS IN NETWORKS: Friendship Paradox & Perception Bias

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DSCI 531

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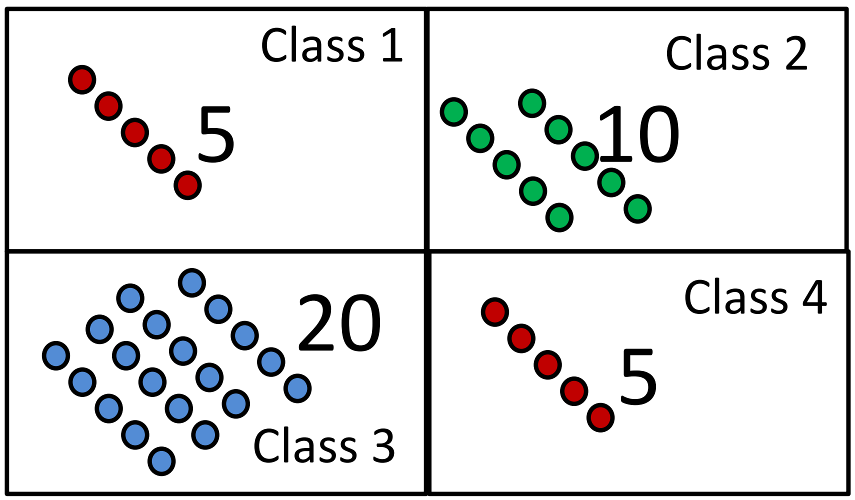
# Class size paradox

- You arrive at a college that boasts average class size of 10 students
- To confirm, you survey students at random about how large their class is
  - 5
  - 5
  - 20
  - 10
  - 5
  - 20
  - …
  - What is the average class size

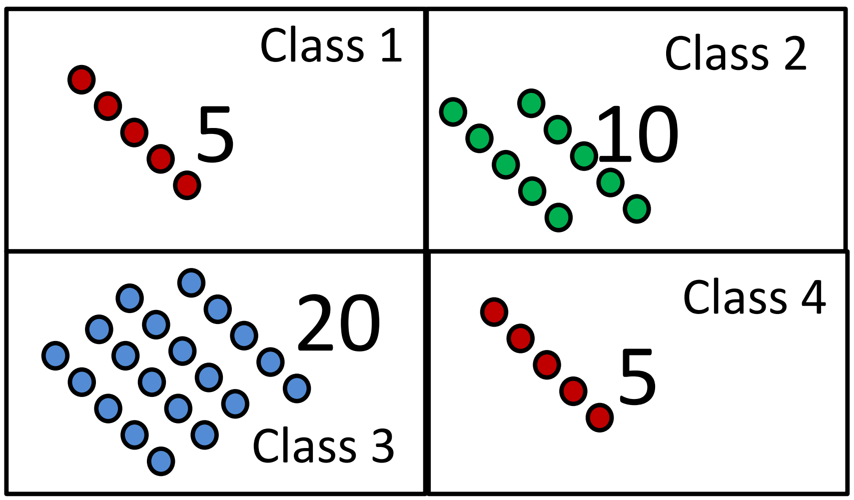
# Class size paradox

- Average class size is …



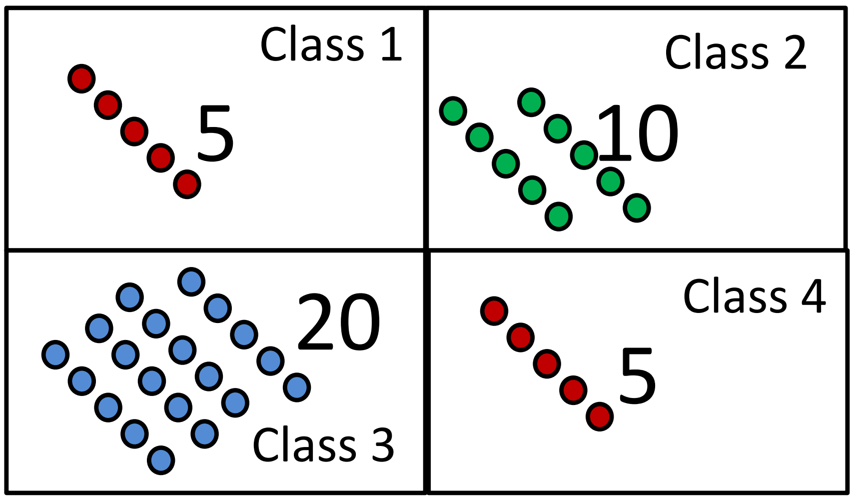
# Class size paradox

- Size of class a student experiences on average is …
  - (average number of classmates)



# Class size paradox

- Average person is associated with larger events than the size of the average event.

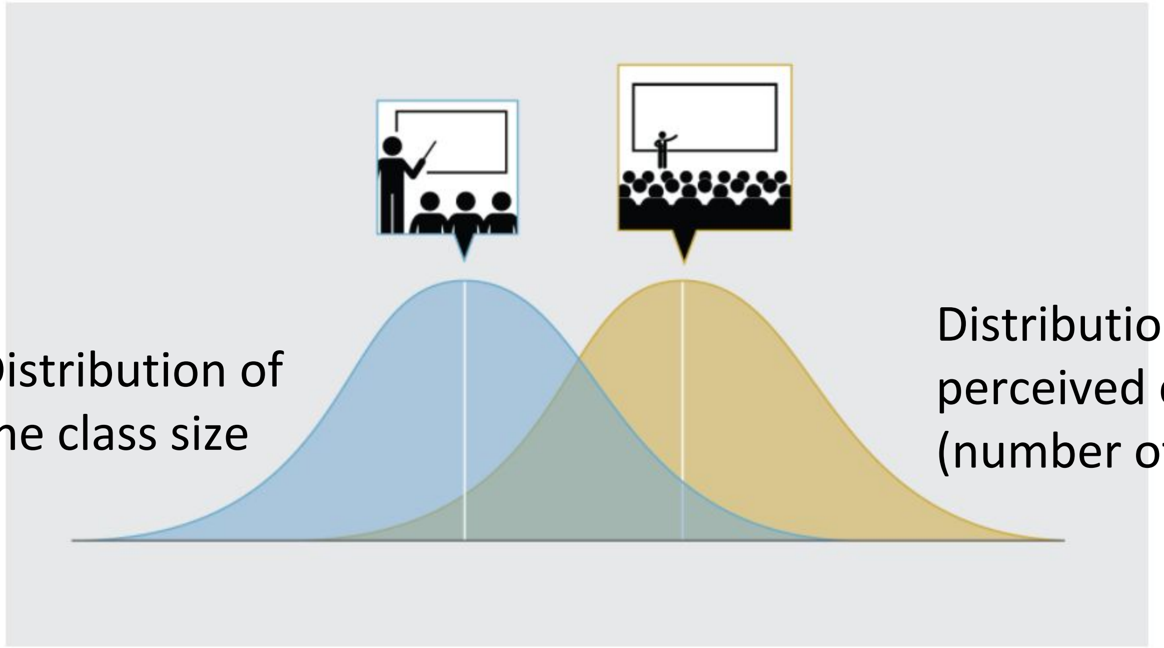



Average size of class is 10

Size of class an average student experiences is 12.75

# Class size paradox

Distribution of the class size



Distribution of the perceived class size (number of classmates)

(Pew Research Center illustration)

# Households are larger at the individual level than at the group level

**Average person's household is larger than average household**

*Household size, by level of analysis*

| | Average individual | Average household |
|---|---|---|
| Sub-Saharan Africa | 6.9 | 5.0 |
| Middle East-North Africa | 6.2 | 5.1 |
| Asia-Pacific | 5.0 | 4.0 |
| Latin America-Caribbean | 4.6 | 3.6 |
| North America | 3.3 | 2.5 |
| Europe | 3.1 | 2.4 |
| World | 4.9 | 3.9 |

Source: Pew Research Center analysis of 2010-2018 census and survey data. See Methodology for details.

- Discrepancies between the average household size and the average experienced by individuals tend to be largest in countries where typical households are large
- The difference between household size as experienced by the average person in the world (4.9 members) is one whole person more than the size of the average household (3.9)

# Differences by household type

**Household types: Distributions change when perspective shifts**

*% in each household type by level of analysis, all countries combined*

| | Asia-Pacific | Europe | Latin America-Caribbean | Middle East-North Africa | North America | Sub-Saharan Africa | World |
|---|---|---|---|---|---|---|---|
| **Individuals** | | | | | | | |
| Extended | 45% | 26% | 32% | 27% | 11% | 35% | 38% |
| Two-parent | 31 | 26 | 39 | 56 | 33 | 37 | 33 |
| Couple | 7 | 19 | 6 | 3 | 20 | 2 | 8 |
| Adult child | 10 | 9 | 10 | 9 | 14 | 2 | 9 |
| Solo | 3 | 13 | 3 | 1 | 11 | 2 | 4 |
| Single-parent | 2 | 4 | 5 | 2 | 9 | 6 | 4 |
| Polygamous | <0.5 | <0.5 | <0.5 | 0.9 | <0.5 | 11 | 2 |
| Other | 1 | 2 | 4 | <0.5 | 1 | 4 | 2 |
| **Households** | | | | | | | |
| Extended | 32% | 17% | 23% | 19% | 8% | 29% | 26% |
| Two-parent | 28 | 16 | 33 | 53 | 19 | 34 | 27 |
| Couple | 14 | 23 | 11 | 8 | 26 | 5 | 15 |
| Adult child | 12 | 8 | 12 | 11 | 12 | 3 | 10 |
| Solo | 11 | 30 | 11 | 5 | 28 | 12 | 15 |
| Single-parent | 3 | 4 | 6 | 3 | 7 | 8 | 4 |
| Polygamous | <0.5 | <0.5 | <0.5 | <0.5 | <0.5 | 5 | 0.5 |
| Other | 0.9 | 2 | 3 | <0.5 | 1 | 3 | 1 |

Source: Pew Research Center analysis of 2010-2018 census and survey data. See Methodology for details.
"Religion and Living Arrangements Around the World"

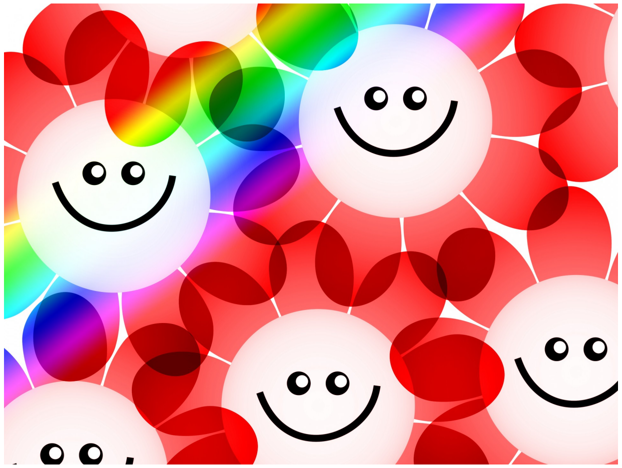
PEW RESEARCH CENTER

- Differences in the shares of people who live in different household *types*.
  - For example, 26% of households in the 130 countries contained extended families, but 38% of people in the same countries live in extended family households, making this the most common living arrangement in the world.
  - Solo households make up 15% of homes globally, but only 4% of people live alone.
- Similar arguments for white people experience shops to be more crowded than they are, and freeways to be more congested than they really are

# This explains why your perceptions are biased

Your friends
are happier

Your friends
drink more

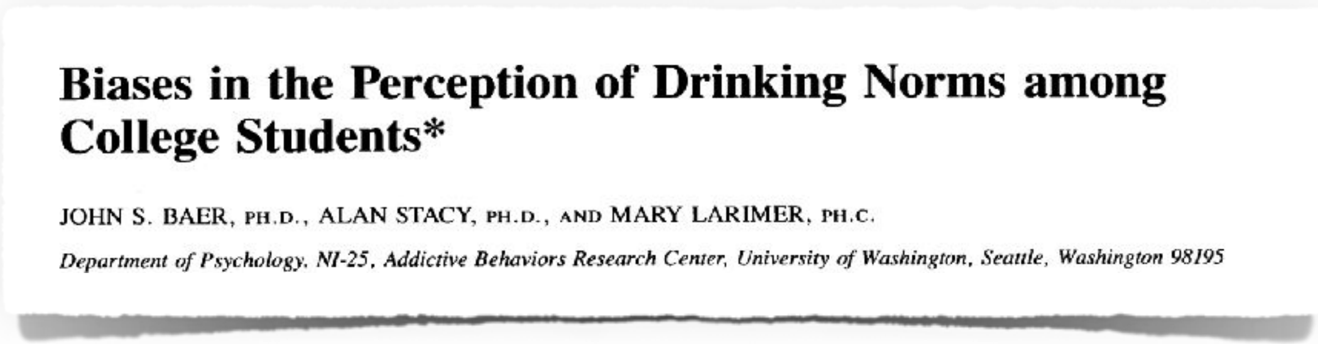


# The problem of binge drinking among (US) students

•10-25% of college students are “heavy" or “problem" drinkers.

This causes issues:

- collegiate adjustment problems
- accidents
- academic failure
- vandalism
- development of mental health disorders

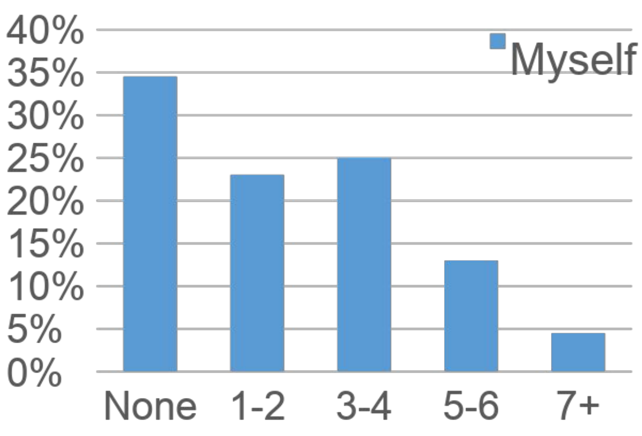


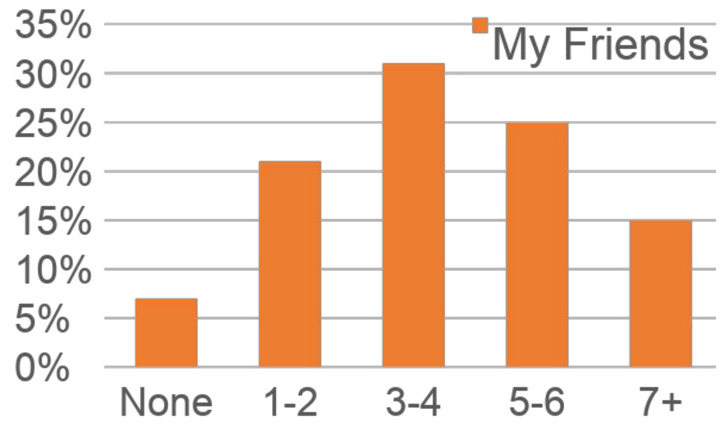
**Biases in the Perception of Drinking Norms among College Students***

JOHN S. BAER, PH.D., ALAN STACY, PH.D., AND MARY LARIMER, PH.C.

*Department of Psychology, NI-25, Addictive Behaviors Research Center, University of Washington, Seattle, Washington 98195*

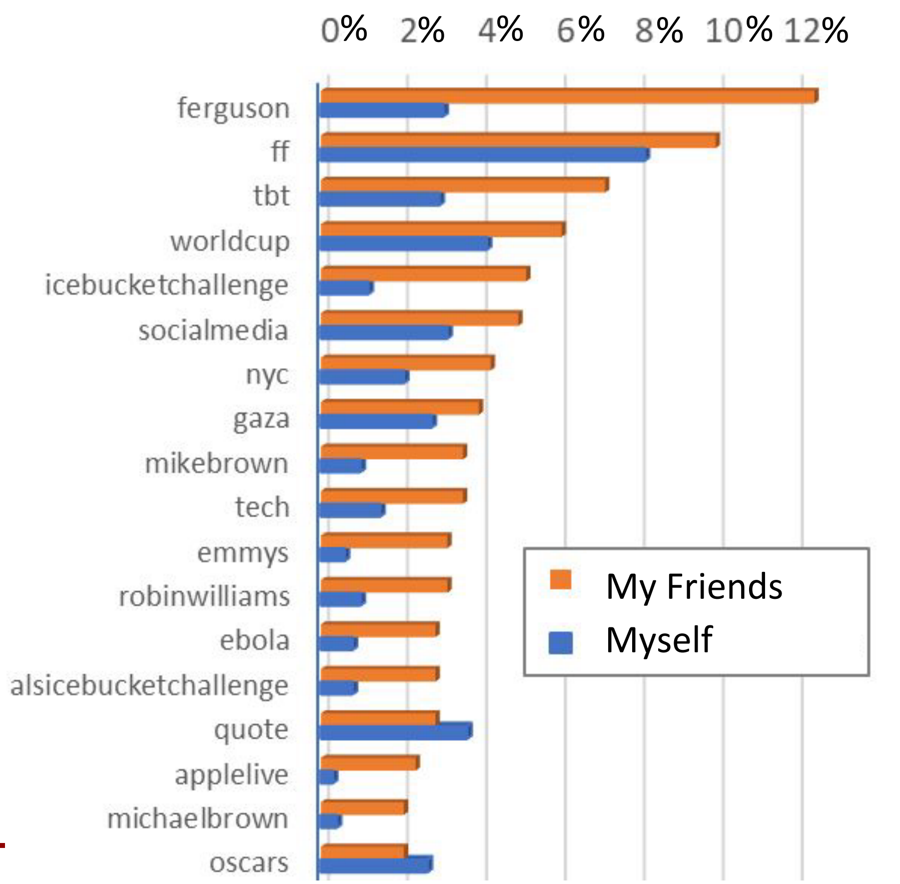
How many alcoholic drinks are consumed at a party?
Virtually all students report that their friends drink more




Source: Most Students Do PartySafe@Cal

# Twitter users overestimate the popularity of some topics




- Compare global to local popularity of topics
  - Myself = Global popularity = fraction of all people talking about a topic
  - My Friends = Local popularity = fraction of friends talking about the topic
- Topics: Popular Twitter hashtags in Summer 2014
- Current events topics appear more popular than they really are

[Alipourfard et al (2020) “Friendship paradox biases perceptions in directed networks”, *Nature Communications*]

# Why?

**The Social-Network Illusion That Tricks Your Mind**

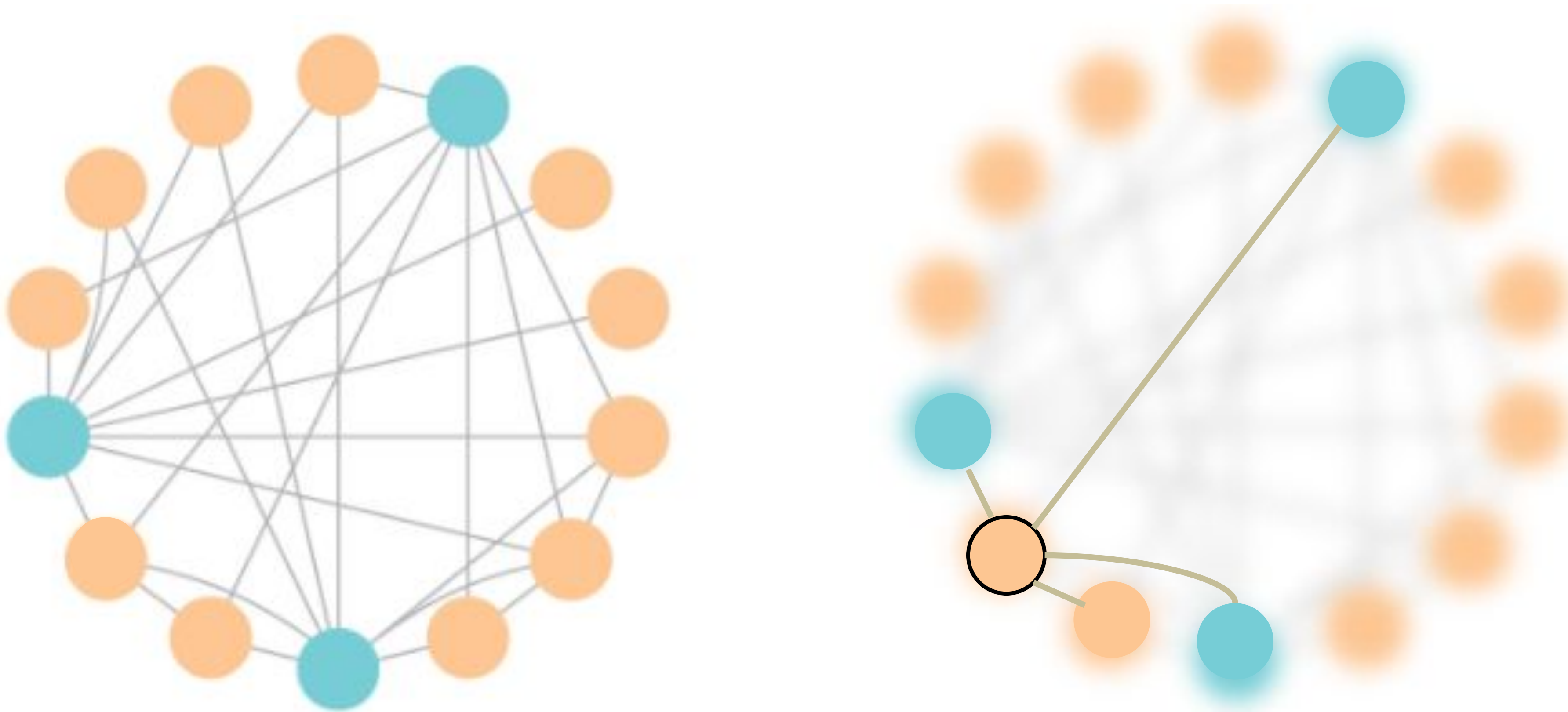
Network scientists have discovered how social networks can create the illusion that something is common when it is actually rare.

June 30, 2015

- Friends are not a random sample of the population!

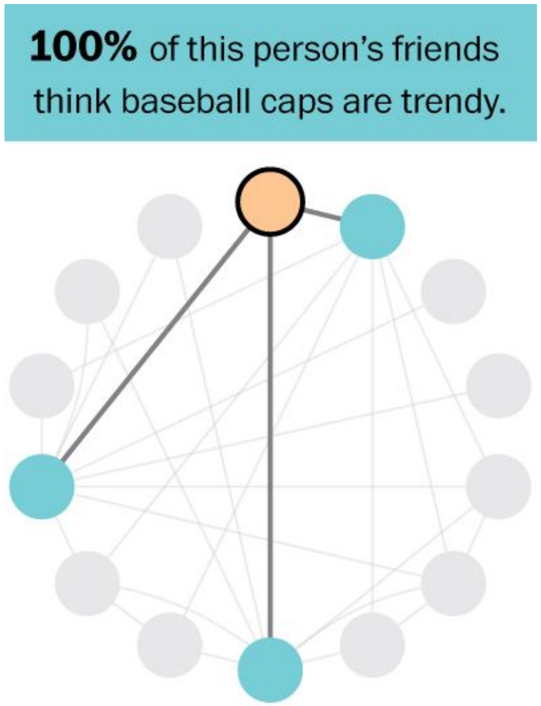
- People seldom see representative sample of the population.
- Instead, their samples — their friends – are highly biased samples of the population
- This, in turn, biases their perception of how common different attributes are in the general population
- These over- and underestimation errors, which we call **perception biases**, affect people's judgements of minority- and majority-group sizes .

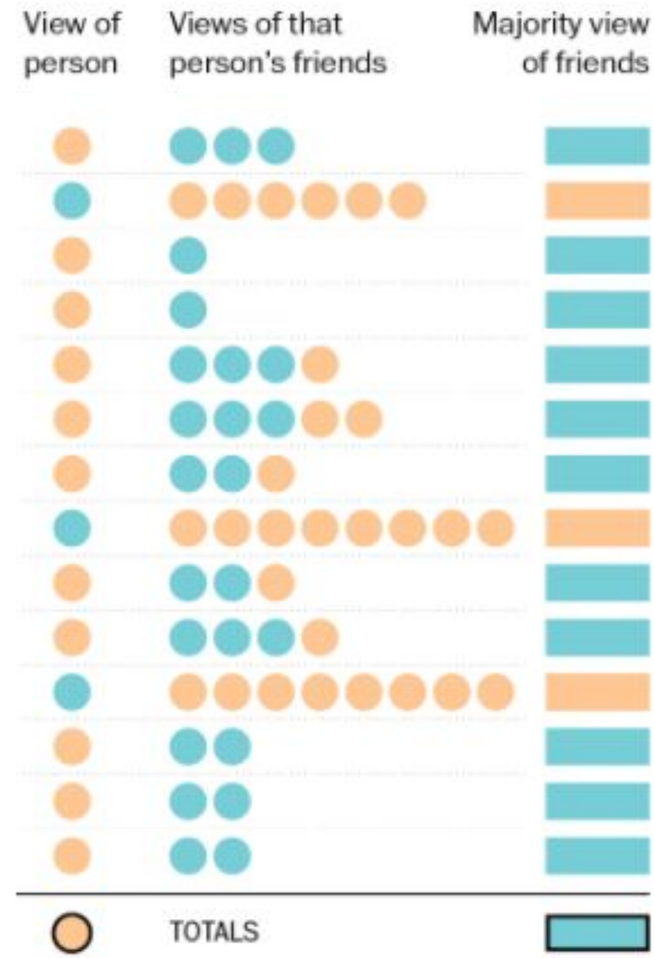
# Social perception bias is a network effect



[Lerman, Wu & Yan (2016) The "Majority Illusion" in Social Networks, in *Plos One*.]

# Local vs global views




K. Schaul, "A quick puzzle to tell whether you know what people are thinking", *Wonkblog, Washington Post*
https://www.washingtonpost.com/graphics/business/wonkblog/majority-illusion/

# FRIENDSHIP PARADOX IN NETWORKS

**Friendship paradox**

Your friends have more friends than you do, on average [Feld, 1991]

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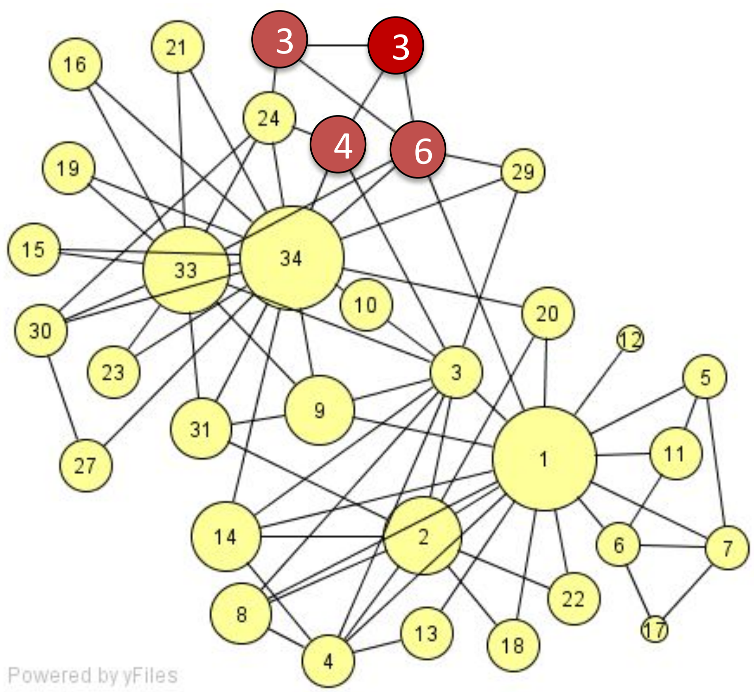

## Friendship paradox

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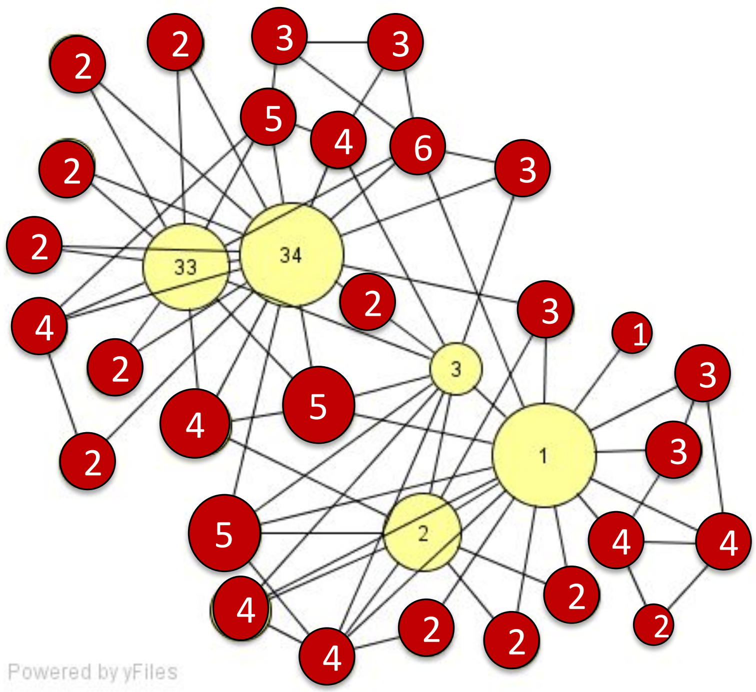
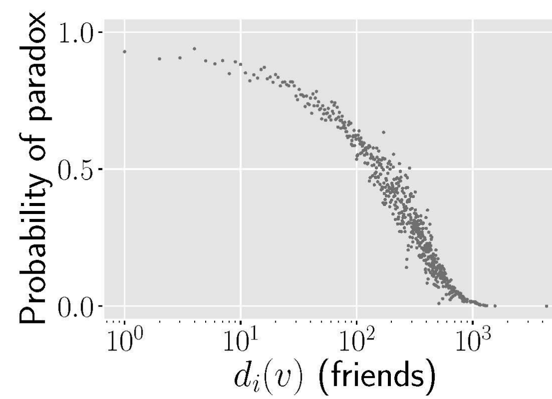


x: e.g. number of friends, wealthiness, productivity, etc.
ρkx: the correlation between the degree k and value x
σk: variance of degree
σx: variance of x

$$F = \langle x \rangle_{nn} - \langle x \rangle = \frac{\rho_{kx}\sigma_k\sigma_x}{\langle k \rangle}.$$

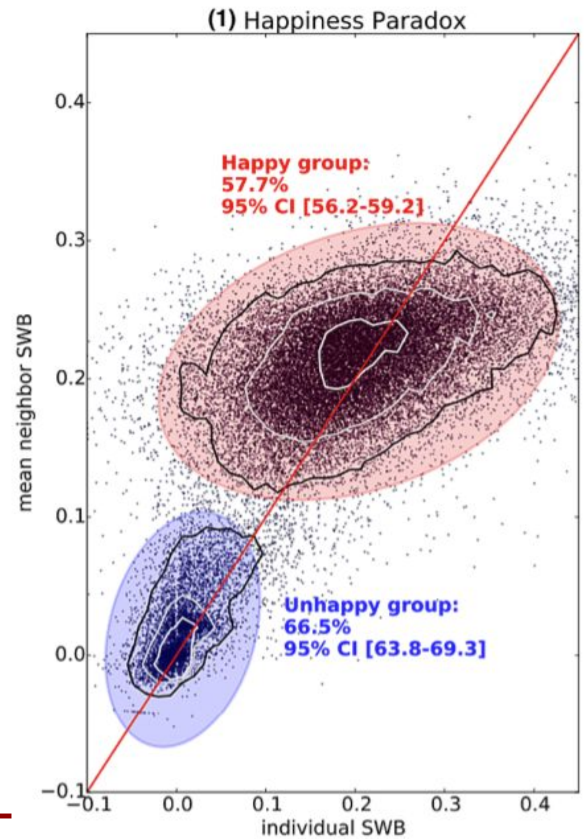
## Friendship paradox

Your friends have more friends than you do, on average [Feld, 1991]

## Generalized friendship paradox

Your friends are more X than you are, on average [Hodas et al., 2013, Eom & Jo, 2014]

- Your friends are happier than you are, on average (Bollen 2017)
- Your coauthors are more productive and more recognized than you are, on average (Eom & Jo 2014)
- Your coauthors have more impact than you do, on average (Benevenuto 2016)
- True whenever more popular people have more **X**



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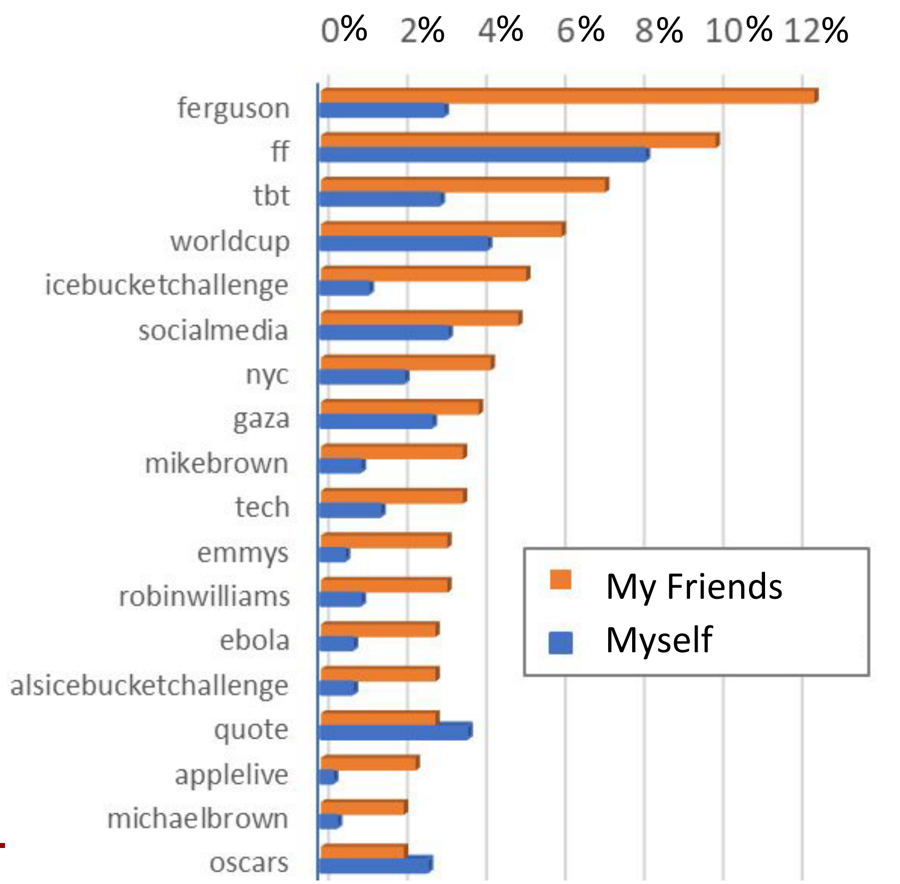
**Perception bias**

Some traits appear more popular within social circles than they really are [Alipourfard et al, 2020]

# Perception bias on Twitter: users overestimate the popularity of some topics

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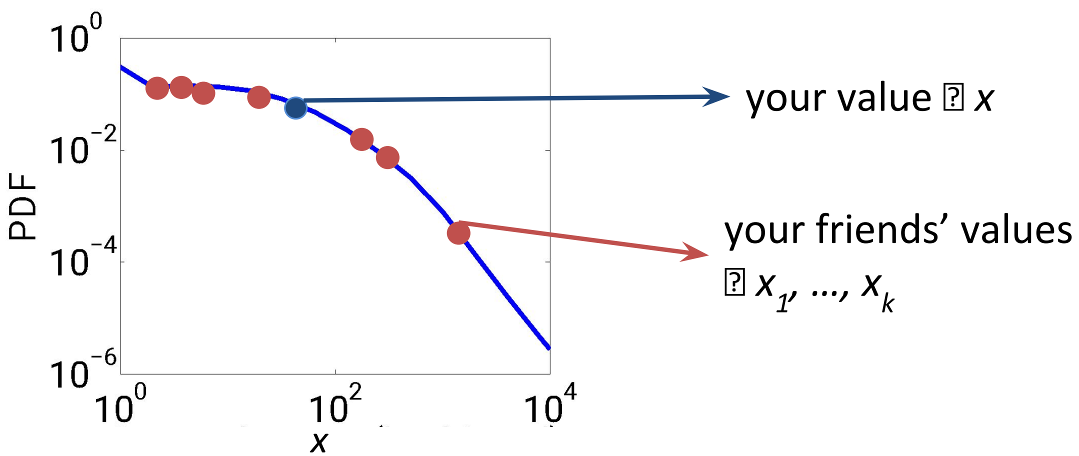
# Friendship paradox as a byproduct of sampling from a heterogeneous distribution



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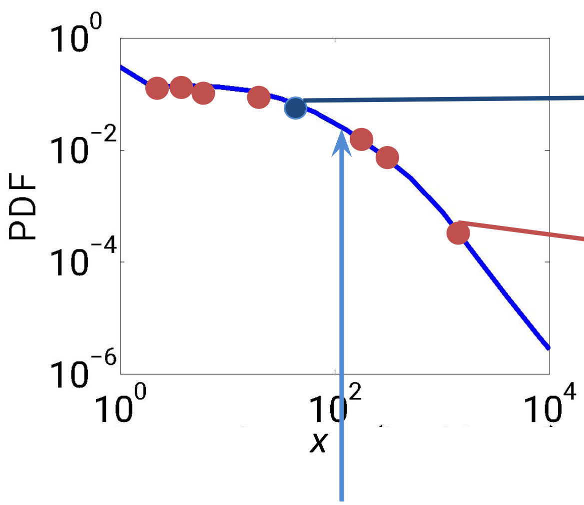


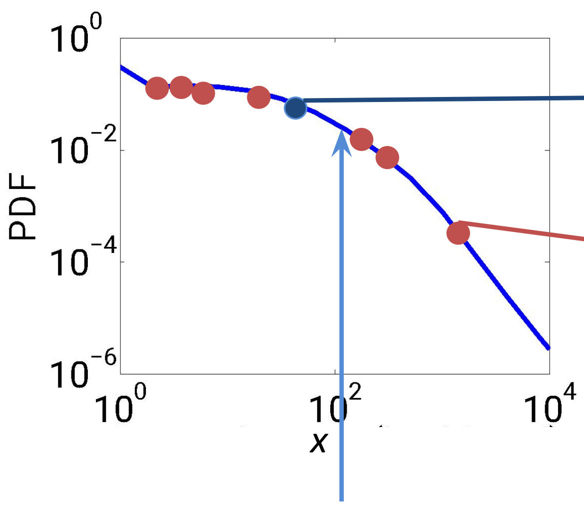

your value ⍰ $x$

your friends' values ⍰ $x_1, \ldots, x_k$

almost always, the sample mean will be larger than the actual x you draw

**Sample mean grows with *k***

# Friendship paradox as a byproduct of sampling from a heterogeneous distribution




your value □ $x$

your friends' values □ $x_1, \ldots, x_k$

**□ use the median instead**

**Friendship paradox**

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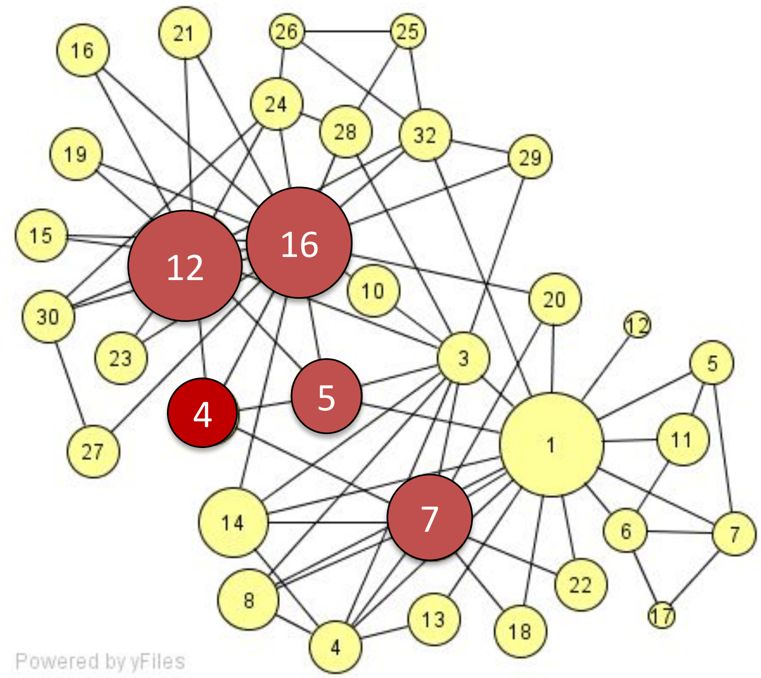
**Strong friendship paradox**

Most of your friends have more friends than you do [Kooti et al., 2014]

# Strong friendship paradox

Most of your friends have more friends than you do

[Kooti, Hodas and Lerman, 2014].



# Strong friendship paradox

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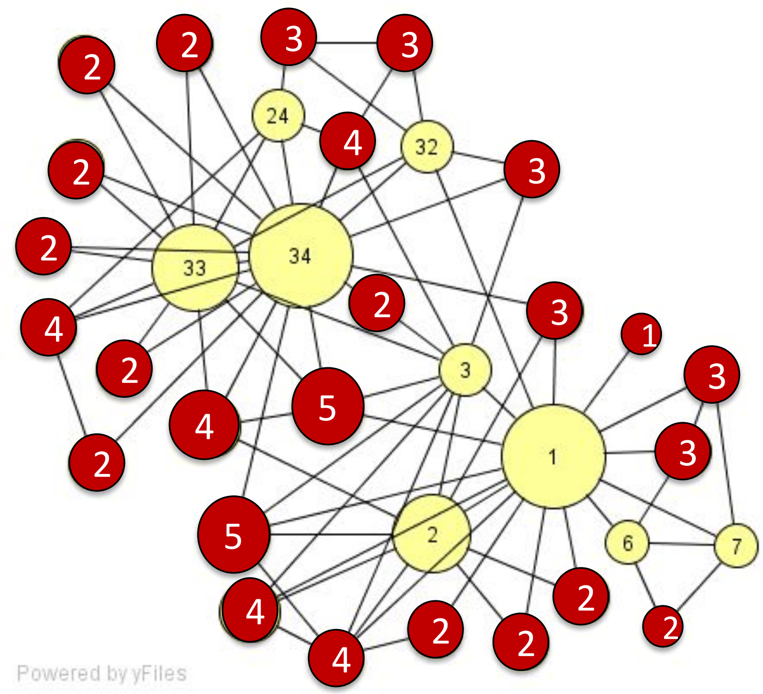



all of the red nodes will experience the paradox

# Friendship paradox is very common…

| Network | Type | Nodes | Probability of paradox |
|---|---|---|---|
| LiveJournal | Social | 3,997,962 | **84%** |
| Twitter | Social | 780,000 | **98%** |
| Skitter | Internet | 1,696,415 | **89%** |
| Google | Hyperlink | 875,713 | **77%** |
| ProsperLoan | Social Finance | 89,269 | **88%** |
| ArXiv | Citation | 34,546 | **79%** |
| WordNet | Semantic | 146,005 | **75%** |

**But wait, there is more…**

**Friendship paradox**

Your friends have more friends than you do, on average [Feld, 1991]

**Generalized friendship paradox**

Your friends are more X than you are, on average [Hodas et al., 2013, Eom & Jo, 2014]

**Perception bias**

Some traits appear more popular within social circles than they really are [Alipourfard et al, 2020]

**Strong friendship paradox**

Most of your friends have more friends than you do [Kooti et al., 2014]

**Generalized strong friendship**

Most of your friends are more X than you are [Kooti et al, 2014]

**Majority illusion**

Most of your friends have a trait, even when it is rare. [Lerman et al, 2016]

## Strong Friendship Paradox

Most of your friends (& followers) are more popular than you are

## Strong Activity Paradox

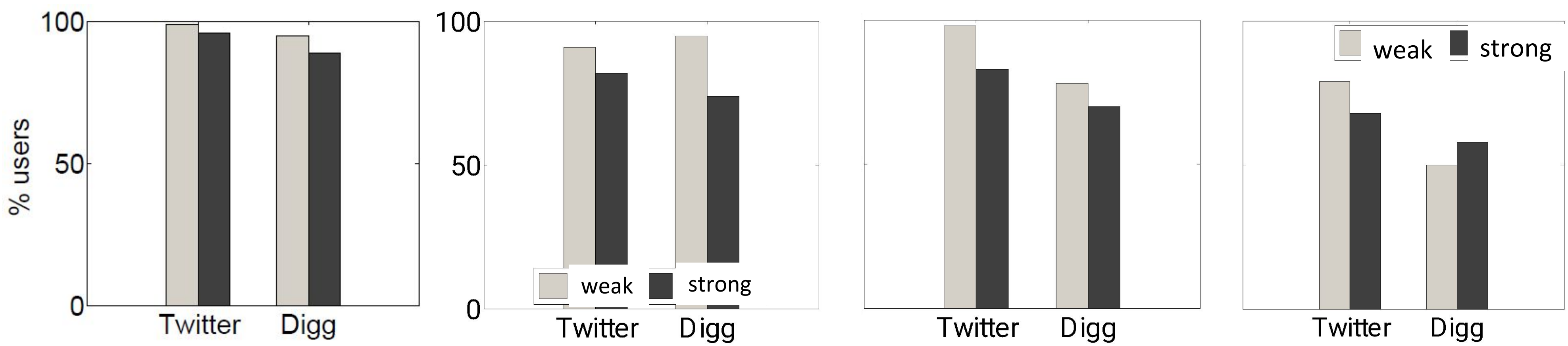
Most of your friends tweet more than you do

## Strong Diversity Paradox

Most of your friends see more diverse content than you do

## Strong Virality Paradox

Most of your friends see more viral content than you do

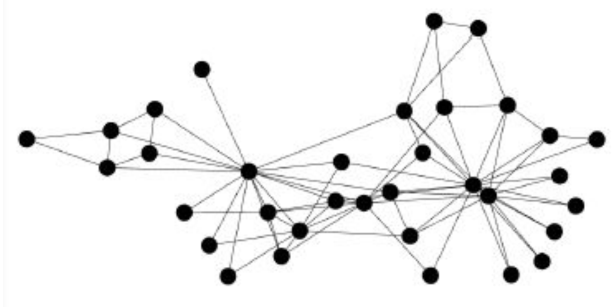


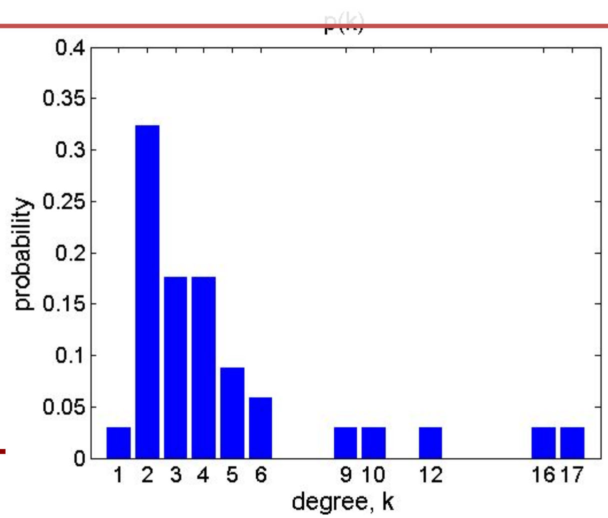

[Kooti, et al (2014) "Network Weirdness: Exploring the origins of network paradoxes" in *ICWSM*]

# QUANTIFYING FRIENDSHIP PARADOX: THE ROLE OF NETWORK STRUCTURE

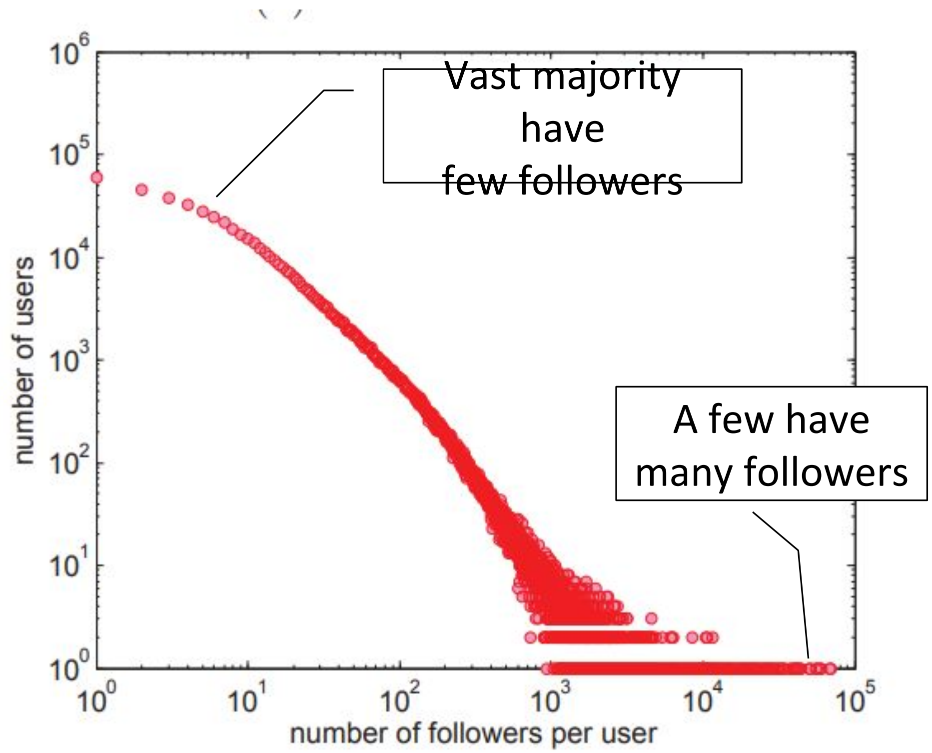
# Degree distribution

- Node degree distribution $p(k)$
  - Probability that a randomly selected ***node*** has degree $k$.
- Explains Feld's friendship paradox
  - Any skewed degree distribution (variance > 0) will lead to a paradox





# Distribution of popularity on Twitter (ca 2010)



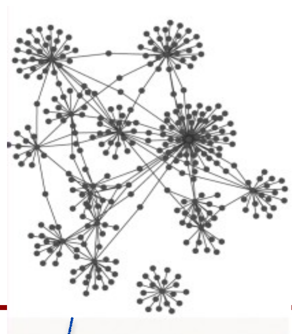
# Assortativity

… Nodes do not link at random

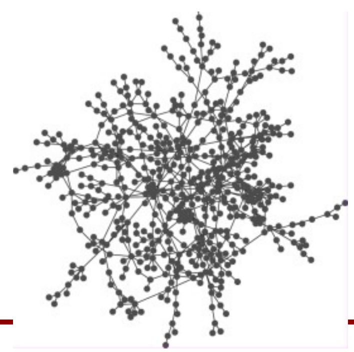
- Joint degree distribution of connected *pairs of nodes* $e(k,k')$
  - Probability that a randomly selected edge links nodes with degrees $k$ and $k'$
  - Explains Feld's friendship paradox, but not strong one
- **Degree assortativity**

MEJ Newman, Assortative Mixing in Networks, *Phys Rev Lett* 89 208701 (2002)

disassortative



assortative







disassortative:
high degree people tend to be
connected to low degree people

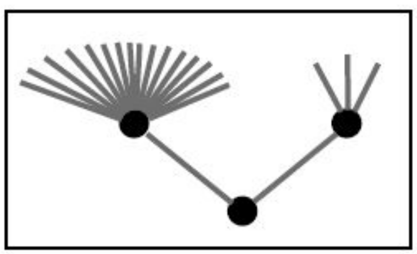
# Transsortativity

… nodes do not link at random

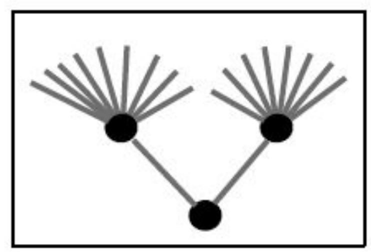
- Joint degree distribution of connected *triplets* of nodes
- **Transsortativity:** neighbors tend to have similar degrees
- Networks can have the same lower-order structure (degree and joint degree distributions) but different higher-order structure

Ngo, Percus, Burghardt & Lerman. The Transsortative Structure of Networks, *in RSPA* (2020)

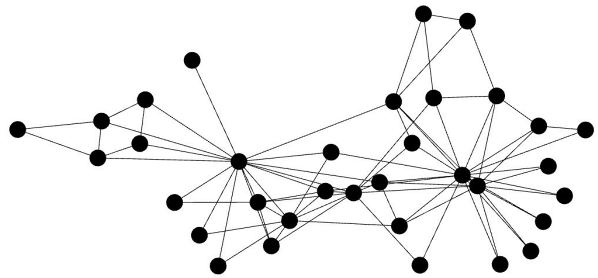
negative transsortativity

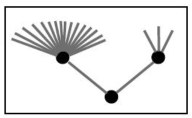


positive transsortativity



# Transsortativity changes network structure





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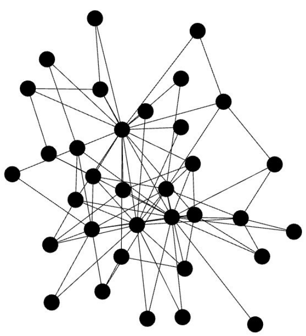
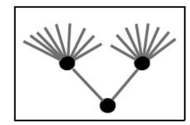


original network



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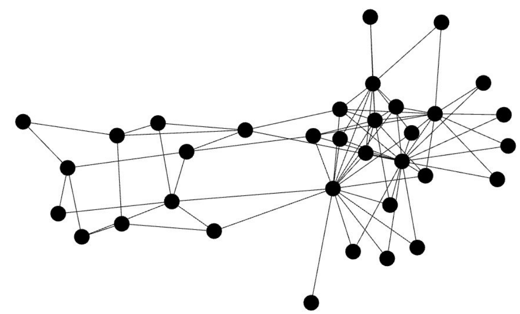
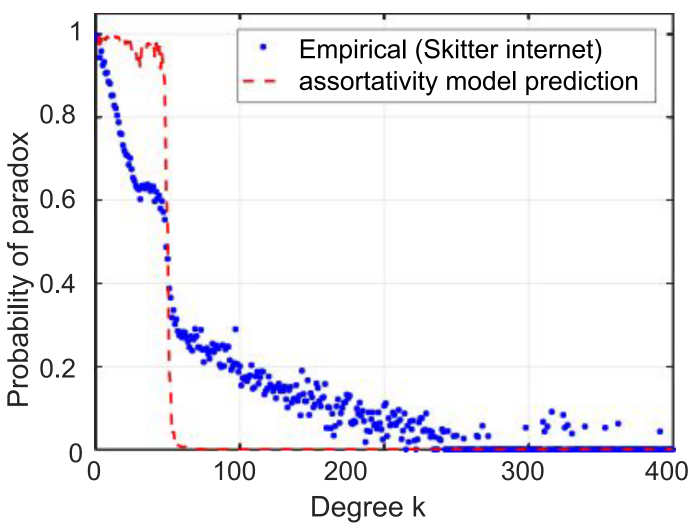


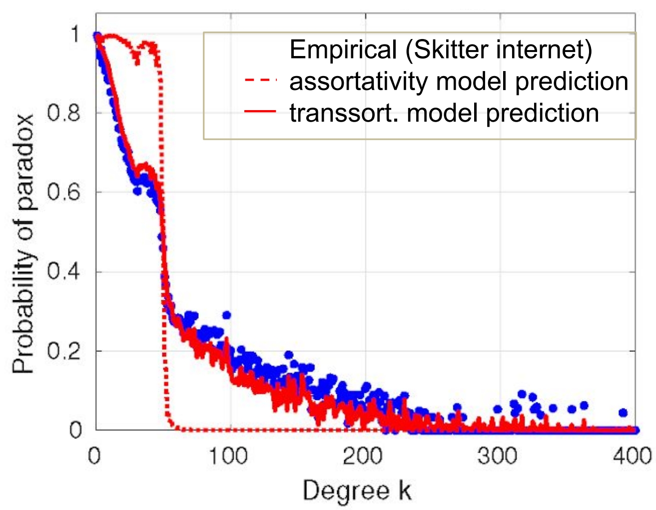
* all networks have the same lower order structure: degree and joint degree distributions

[Ngo, Percus, Burghardt & Lerman (2020) The Transsortative Structure of Networks, *in J Royal Society Proc. A*]

# Transsortativity explains strong friendship paradox

Given a network, calculate the fraction of degree-*k* nodes the majority of whose neighbors have a higher degree



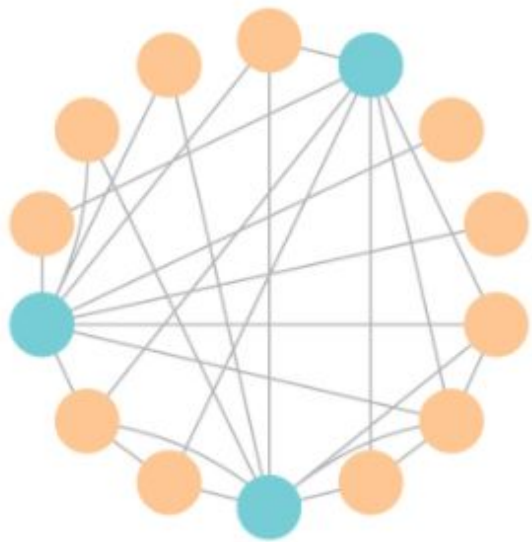




* transsortative structure is sufficient to explain the paradox

[Wu et al (2017) "Neighbor-neighbor correlations explain measurement bias in networks" *Scientific Reports* 7]

# From friendship paradox to “majority Illusion”

- Nodes have a trait: active/not, yellow/blue, heavy drinker/teetotaler, …
- Trait can be rare, but many nodes will overestimate its prevalence



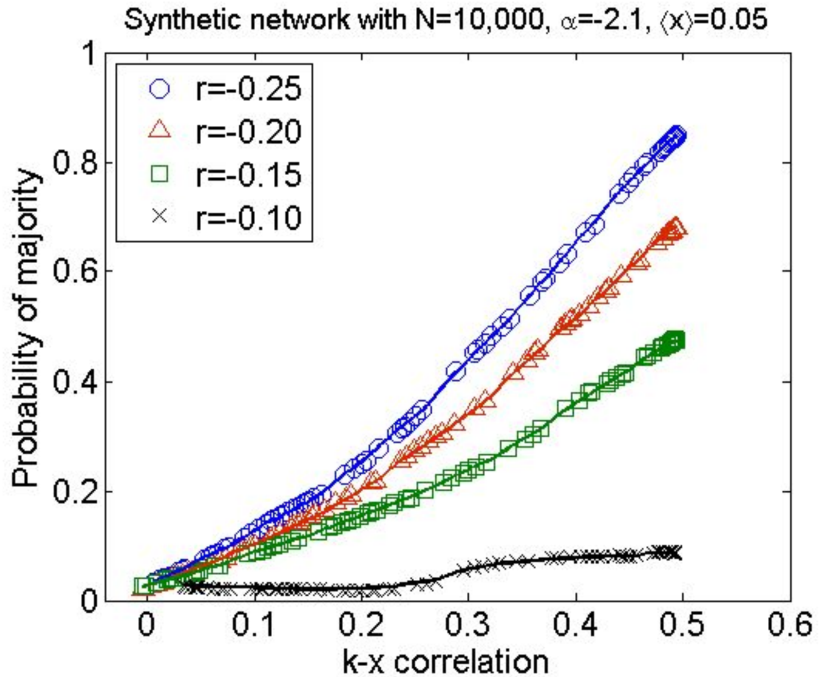
[Lerman, Wu & Yan (2016) The “Majority Illusion” in Social Networks, in *Plos One.*]

# Network structure & majority illusion

•Network structure can amplify the majority illusion when:

- **Influential (high degree) nodes are likely to have the trait:** k-x correlation measures likelihood high-degree nodes have the trait
- **Skewed degree distribution:** few high-degree nodes amplify the effect
- **Disassortativity:** High degree nodes link to low degree nodes: negative assortativity ($r_{2k}$<0)

Synthetic network with 5% blue nodes




[Lerman, Wu & Yan (2016) The "Majority Illusion" in Social Networks, in *Plos One.*]

# Inequalities in social networks

Lee, E., Karimi, F., Wagner, C., Jo, H. H., Strohmaier, M., & Galesic, M. (2019). Homophily and minority-group size explain perception biases in social networks. *Nature Human Behaviour*, 1-10.

**1.) Homophily influences ranking of minorities**

Minorities are more affected by homophilic interactions and their group size than majorities

**2.) Homophily can impact perception of groups**

Evidence for perception biases on social network and population levels, can be (partially) mitigated



下一页是解释

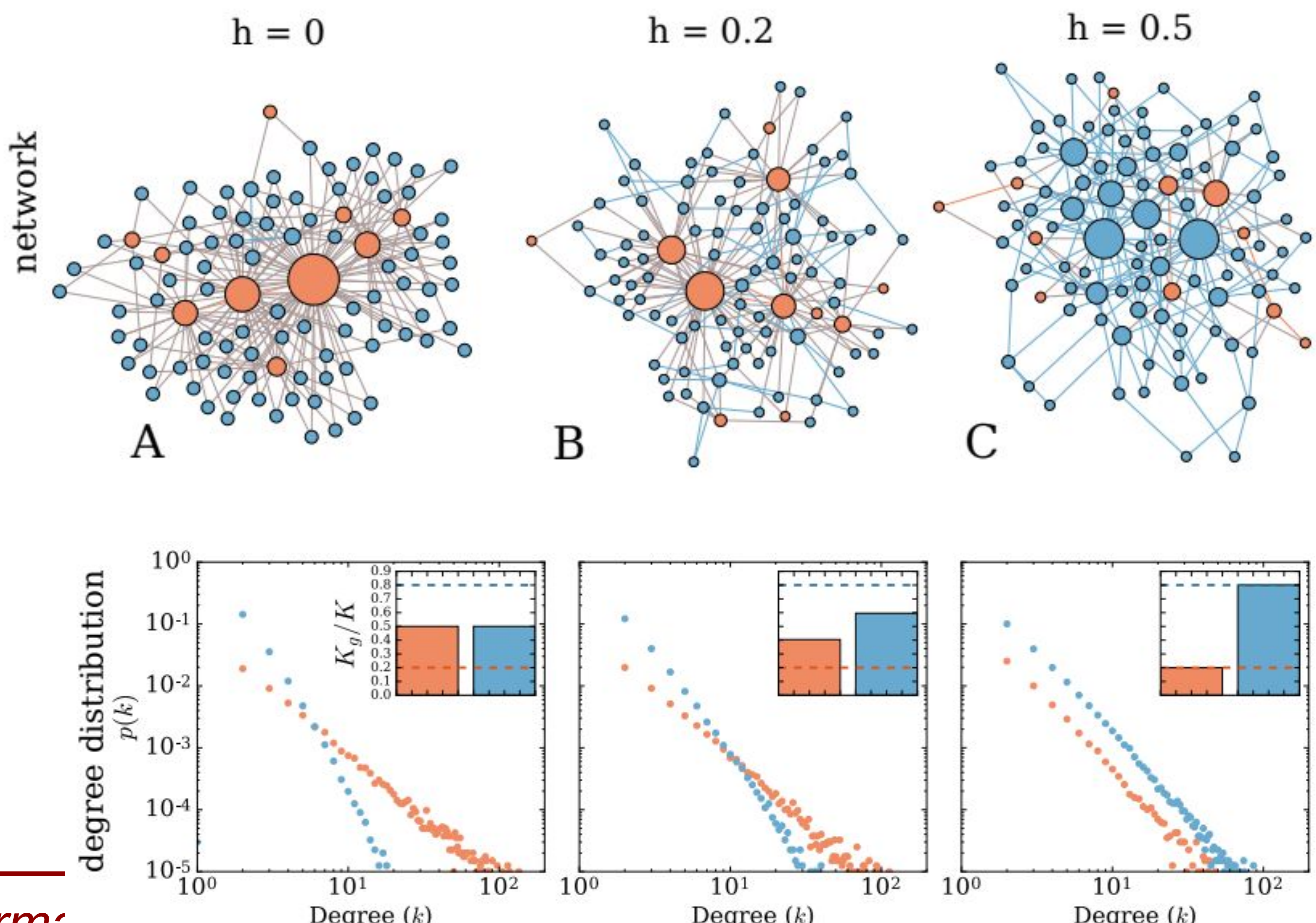
## 1.) Homophily influences ranking of minorities

- **Key Idea:** Minority groups experience stronger effects from homophily compared to majority groups.
- **Why?** In social networks, if people prefer to connect with others similar to them (homophily), then minorities have fewer same-group connections simply due to their smaller numbers. This can affect their visibility and status within the network.
- **Ranking Effect:** Since network centrality (importance) is often determined by connections, minorities might be ranked lower in social influence compared to the majority, even if they have equal capabilities.

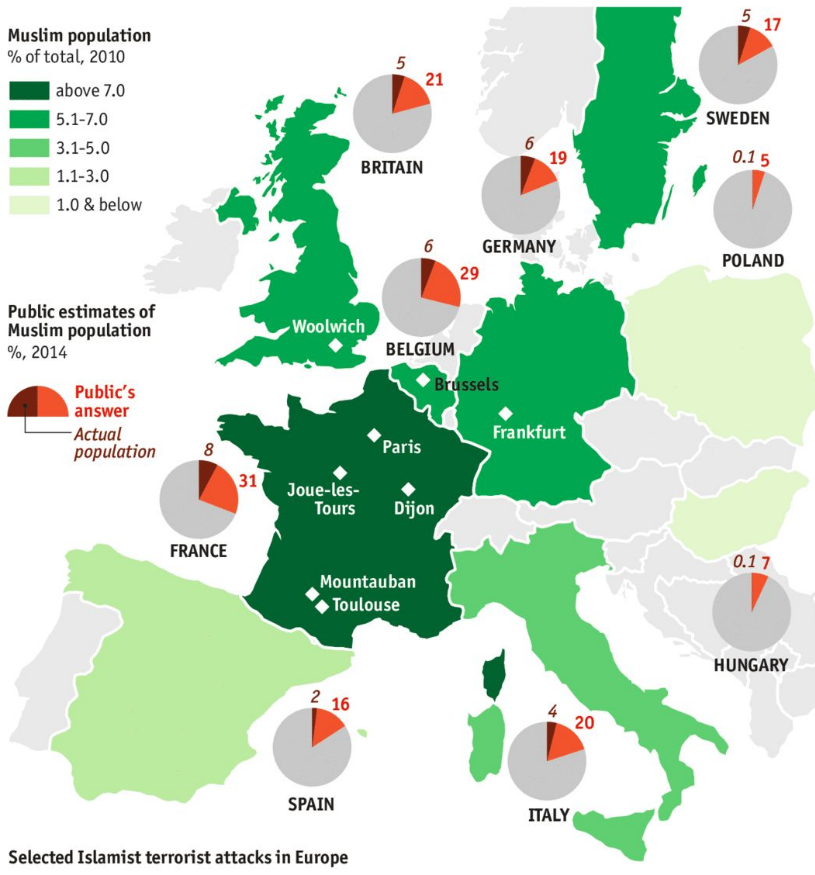
## 2.) Homophily can impact perception of groups

- **Key Idea:** Homophily can lead to biases in how groups are perceived at both individual and network levels.
- **How?** If people primarily interact within their own group, they may develop an inaccurate perception of the overall population, overestimating the size and influence of their own group while underestimating minorities.
- **Mitigation:** The paper suggests that while homophily leads to these biases, they can be (partially) reduced through structural changes in the network, such as increasing diverse connections or exposure to different groups.

# Ranking bias in networks



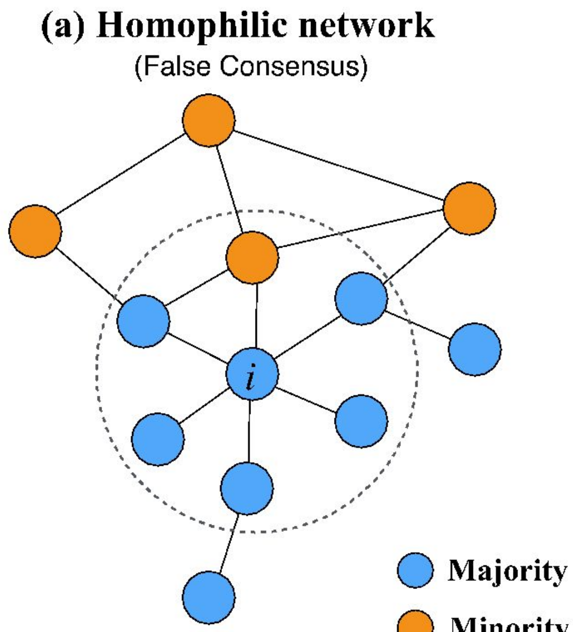
# Perception of minorities



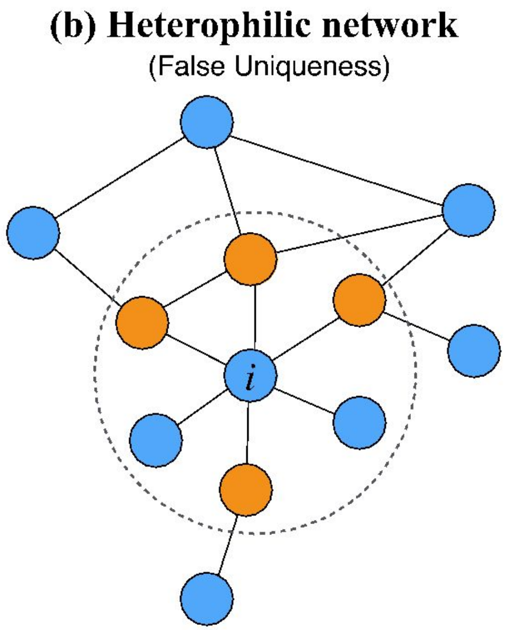

perception > actual
overestimating

Slide courtesy of Markus Strohmaier

# Perception of minorities in social networks

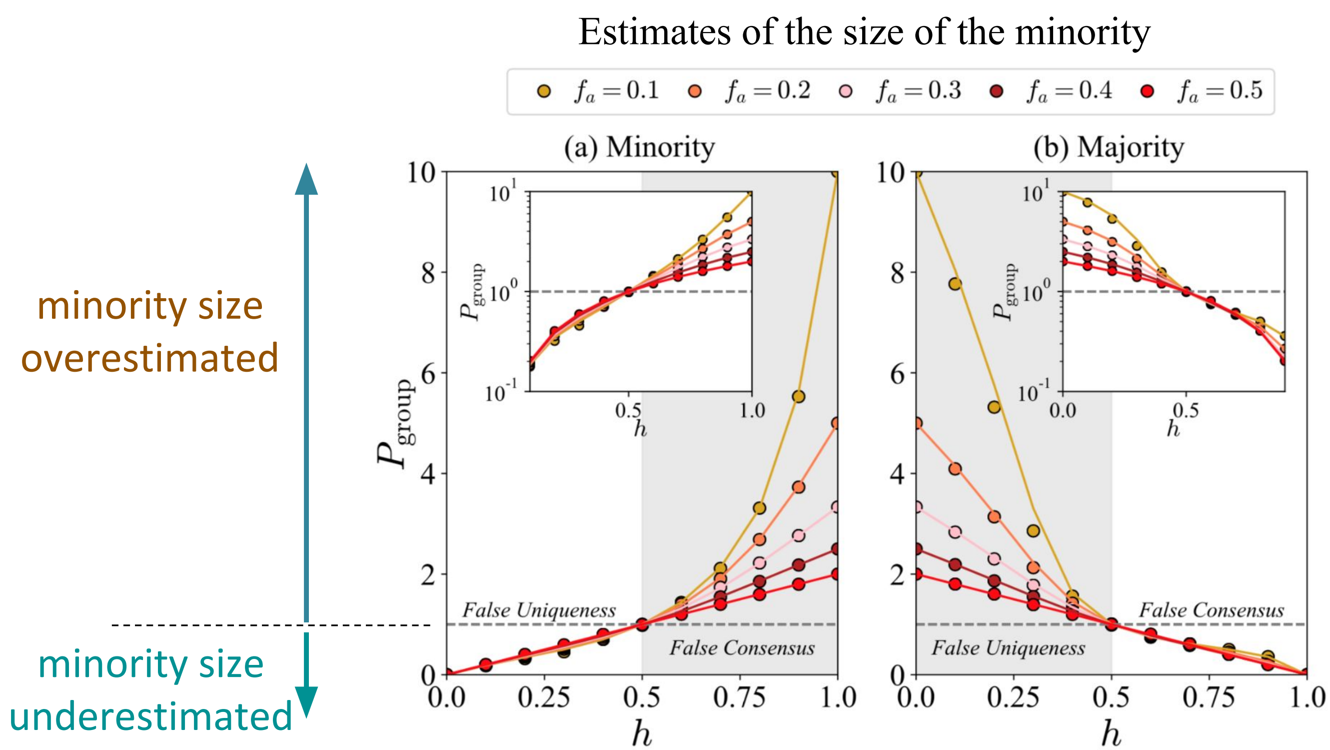



$P_{indiv}(i) = 1/6 \approx 16\%$
(0.5 fold underestimation)



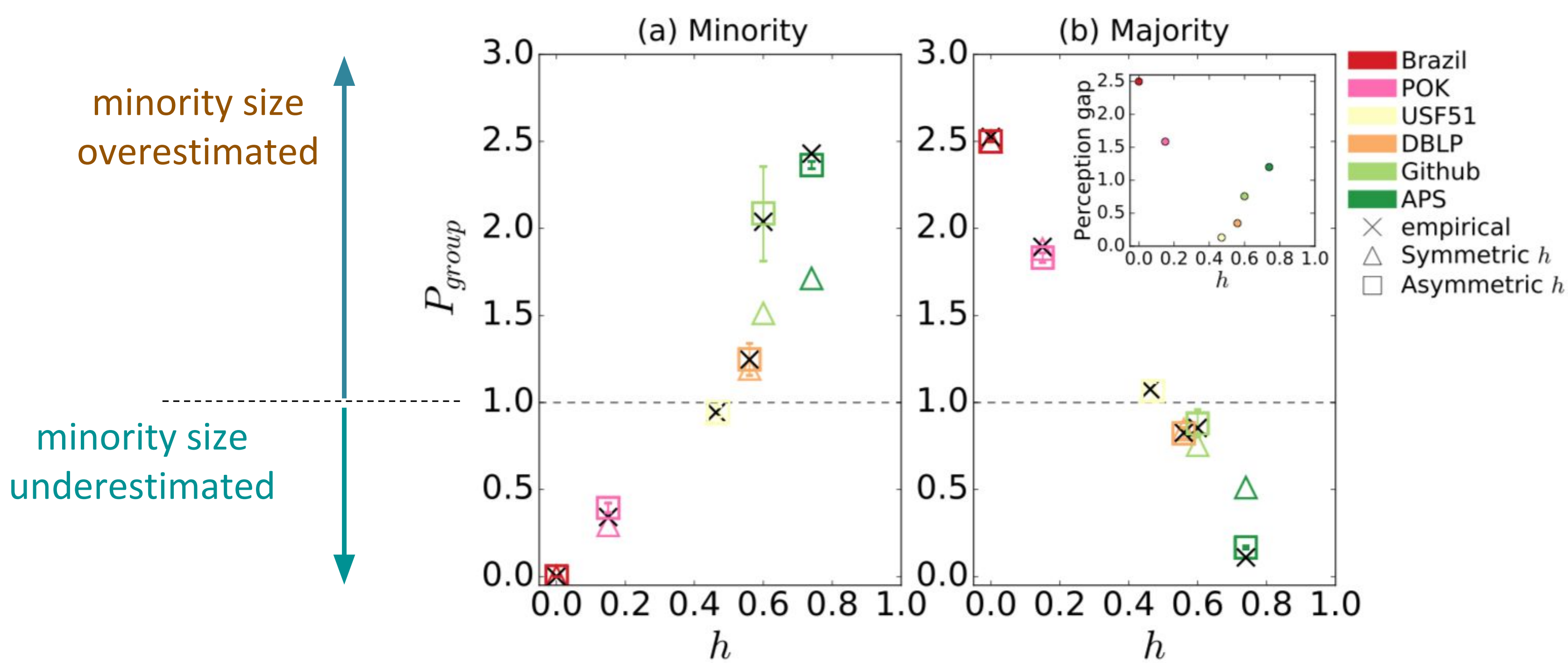

$P_{indiv}(i) = 4/6 \approx 67\%$
(2 fold overestimation)

# Perception bias vs. homophily and minority size



# Perception bias in empirical data

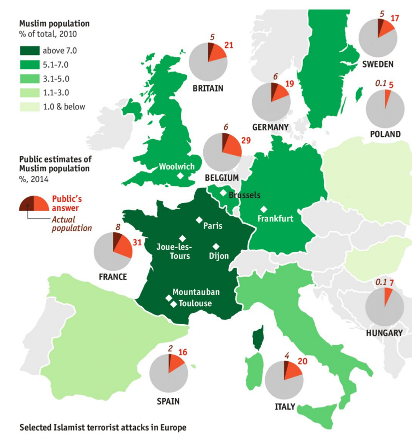
Estimates of the size of the minority



# Can we observe similar perception biases on a national level?

- International survey programs including the **US**, **Germany** and **South Korea**:

| Characteristic | Question text | Original source of the question text the data for the general population | | |
|---|---|---|---|---|
| | | US | Germany | S.Korea |
| 1.Not having money for food | Have there been times in the past 12 months when you did not have enough money to buy food you or your family needed? Yes–No | Gallup World Poll (2010) | Gallup World Poll (2010) | Social Integration Status Survey (2011) |
| 2. Donating to charity | In the past month, have you donated money to a charity? Yes–No | Gallup World Poll (2010) | Gallup World Poll (2010) | Social Survey of Welfare (2017) |
| 3. Experiencing theft | Within the past 12 months, have you had money or property stolen from you or another household member? Yes–No | Gallup World Poll (2010) | Gallup World Poll (2010) | - |
| 4. Religion importance | Is religion an important part of your daily life? Yes–No | Gallup World Poll (2010) | Gallup World Poll (2010) | Religion of Koreans by Gallup (1984-2014) |
| 5.Worship attendance | Have you attended a place of worship or a religious service within the past seven days? Yes–No | Gallup World Poll (2010) | Gallup World Poll (2010) | Religion of Koreans by Gallup (1984-2014) |



# Can we observe similar perception biases on a national level?

- Examples of different minority / majority issues:

| Characteristics | US(%) | | Germany(%) | | Korea(%) | |
|---|---|---|---|---|---|---|
| | Yes | No | Yes | No | Yes | No |
| 1. Not having money for food | 19 | 81 | 5 | 95 | 3 | 97 |
| 2. Donating to charity | 57 | 43 | 43 | 57 | 26.7 | 73.7 |
| 3. Experiencing theft | 12 | 88 | 9 | 91 | - | - |
| 4. Religion importance | 70 | 30 | 27 | 73 | 52 | 48 |
| 5. Worship attendance | 53 | 47 | 33 | 67 | 44 | 56 |
| 6. God and morality | 47 | 53 | 33 | 67 | - | - |
| 7. Belief in a god | 64 | 36 | 38 | 62 | 39 | 61 |
| 8. Smoking | 15.2 | 85 | 21.9 | 78 | 23.9 | 76.1 |
| 9. Military force | 76.5 | 24 | 50 | 50 | - | - |
| 10. Homosexuality | 35.5 | 64 | 12.1 | 87.9 | 34 | 58 |

**Table S2.** The object ratio of population for US, Germany, and Korea with the survey questions.

**+ a crowdworker survey** (n=300) asking:

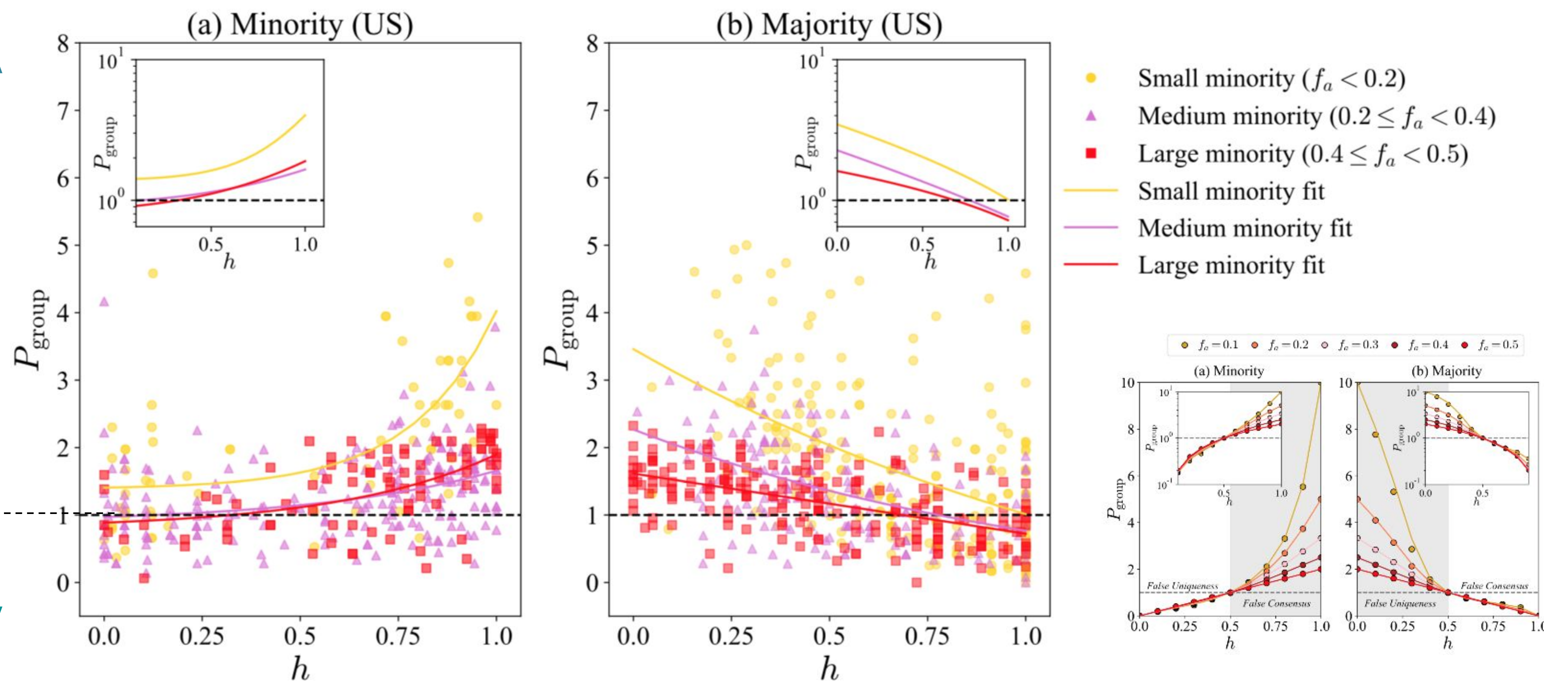
1. Do you have **characteristic *x***?
2. How frequent is **characteristic *x*** in your personal network?
3. How frequent is **characteristic *x*** in the population of your country?

# Perception of minorities on a population level

- Results for minority/majority estimates from the US (results similiar for Germany):

minority size overestimated

minority size underestimated



# Summary

- Friends are not a representative sample of the population
- As a result, your observations/perceptions will be systematically biased
  - E.g., a rare attribute appears far more common than it is
- Impacts
  - Ranking/visibility of minorities in networks
  - Link and content recommendation in networks
  - Psychological well-being
    - Upward social comparison
      - Loneliness and Facebook use: the role of social comparison and rumination (PMID: 33521361)